

§15.3 方向导数和梯度

一. 方向导数

定义3.1. (方向导数). 设 $z = f(x, y)$ 定义于 $E \subset \mathbb{R}^2$. 线段

$[P_0, P_1] \subset E$. L 是从 P_0 出发过 P_1 的

射线. $\vec{l}$ 是 L 方向的单位向量,

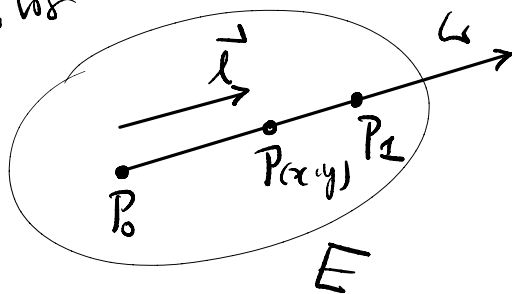
(即 $\overrightarrow{P_0 P_1}$ 与 $\vec{l}$ 同方向)

$\vec{l} = \{\cos \alpha, \cos \beta\}$, α, β 是 $\vec{l}$ 的

方向角 (即 α, β 是 $\vec{l}$ 与 x 轴, y 轴夹角, $\cos \alpha, \cos \beta$

是 $\vec{l}$ 的方向余弦). L 上一点参数化, $P(x, y) \in L$:

$$\begin{cases} x = x(t) = x_0 + t \cos \alpha \\ y = y(t) = y_0 + t \cos \beta \end{cases} \quad t = |\overrightarrow{P_0 P}|.$$



增量 $\Delta f = f(P) - f(P_0)$ — P_0 的邻点 P 与 P_0 函数值之差

$$\frac{\Delta f}{|P_0 P|} = \frac{\Delta f}{t}: \text{在 } P_0 \text{ 点沿 } \vec{l} \text{ 在 } [P_0, P] \text{ 上平均变化率}$$

若 $\lim_{t \rightarrow 0} \frac{\Delta f}{t} = A$: 则称 A 是 $f(x, y)$ 在 P_0 处沿 $\vec{l}$ 的方向

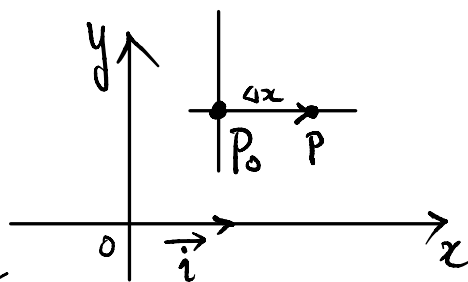
导数. 记为 $\frac{\partial f}{\partial l} \Big|_{P_0} = \frac{\partial f(P_0)}{\partial l} = \frac{\partial f(x_0, y_0)}{\partial l}$.

注: ① 若 $\vec{l} = e_1 = \vec{i} = \{1, 0\}$

$$x = x_0 + t, y = y_0$$

$$\lim_{t \rightarrow 0} \frac{\Delta f}{t} = \lim_{t \rightarrow 0} \frac{f(x_0 + t, y_0) - f(x_0, y_0)}{t}$$

$$= \frac{\partial f}{\partial x} \Big|_{P_0}, \text{ 即 } y_0 \text{ 处 } f \text{ 对 } x \text{ 的偏导数.}$$



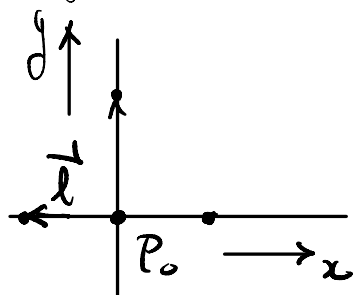
若 $\vec{l} = \vec{e}_2 = \vec{j} = \{0, 1\}$,

$$\lim_{t \rightarrow 0} \frac{\Delta f}{t} = \lim_{t \rightarrow 0} \frac{f(x_0, y_0 + t) - f(x_0, y_0)}{t} = \left. \frac{\partial f}{\partial y} \right|_{P_0} \quad \text{—— 沿 } y \text{ 轴的偏导数}$$

如 $\vec{l} = -\vec{i} = \{-1, 0\}$

$$\frac{\Delta f}{t} = \lim_{t \rightarrow 0} \frac{f(x_0 - t, y_0) - f(x_0, y_0)}{t}$$

$$\stackrel{-t=s}{=} - \lim_{s \rightarrow 0} \frac{f(x_0 + s, y_0) - f(x_0, y_0)}{s} = - \left. \frac{\partial f}{\partial x} \right|_{P_0}$$



② $\vec{l}_1 = -\vec{l} = \{\cos \alpha, \cos \beta\}$.



令 $x = x_0 - t \cos \alpha, \quad y = y_0 - t \cos \beta$

$$\Delta f = f(x, y) - f(x_0, y_0) = f(x_0 - t \cos \alpha, y_0 - t \cos \beta) - f(x_0, y_0)$$

$$\left. \frac{\partial f}{\partial l} \right|_{P_0} = \lim_{t \rightarrow 0} \frac{\Delta f}{t} = \lim_{t \rightarrow 0} \frac{f(x_0 - t \cos \alpha, y_0 - t \cos \beta) - f(x_0, y_0)}{t}$$

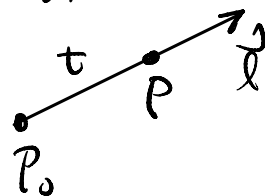
$$\stackrel{t=-s}{=} - \lim_{s \rightarrow 0} \frac{f(x_0 + s \cos \alpha, y_0 + s \cos \beta) - f(x_0, y_0)}{s}$$

$$= - \left. \frac{\partial f}{\partial \vec{l}} \right|_{P_0}$$

(2). 如 $f(x, y)$ 可微, $P_0(x_0, y_0)$ 处: $\vec{l} = \{\cos \alpha, \cos \beta\}$.

$\varphi(t) \triangleq f(x_0 + t \cos \alpha, y_0 + t \cos \beta) - f(x_0, y_0) = \Delta f$.

$$\left. \frac{\partial f}{\partial \vec{l}} \right|_{P_0} = \lim_{t \rightarrow 0} \frac{\Delta f}{t} = \lim_{t \rightarrow 0} \frac{\varphi(t) - \varphi(0)}{t} = \varphi'(0)$$



$$\begin{aligned}
&= f_x \Big|_{t=0} \cdot \cos \alpha + f_y \Big|_{t=0} \cdot \cos \beta \\
&= f_x(x_0, y_0) \cos \alpha + f_y(x_0, y_0) \cdot \cos \beta \\
&= f_x(P_0) \cdot \cos \alpha + f_y(P_0) \cdot \cos \beta \\
&= \{f_x(P_0), f_y(P_0)\} \cdot \{\cos \alpha, \cos \beta\} \\
&= \{f_x(P_0), f_y(P_0)\} \cdot \vec{l} \quad (\because |\vec{l}|=1)
\end{aligned}$$

$$\therefore \frac{\partial f}{\partial \lambda} \Big|_{P_0} = \{f_x(P_0), f_y(P_0)\} \cdot \vec{l}$$

三元函数: $u=f(x, y, z)$ 定义于 $E \subset \mathbb{R}^3$. $[P_0, P_1] \subset E$.

L 是 P_0 到 P_1 的射线. $\vec{l}$ 是 L 上方向. $|\vec{l}|=1$.

$$\vec{l} = \{\cos \alpha, \cos \beta, \cos \gamma\}$$

$\forall P(x, y, z) \in L$:

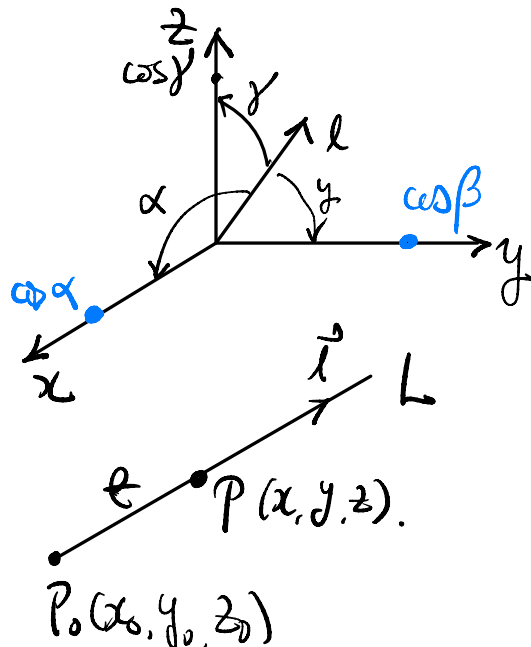
$$\begin{cases} x = x_0 + t \cos \alpha \\ y = y_0 + t \cos \beta \\ z = z_0 + t \cos \gamma \end{cases} \quad t = |\vec{P_0 P}|$$

$$\Delta f = f(x, y, z) - f(x_0, y_0, z_0)$$

$$\lim_{t \rightarrow 0} \frac{\Delta f}{t} = \lim_{P \rightarrow P_0} \frac{f(P) - f(P_0)}{|\vec{P_0 P}|} = A.$$

A 是 $f(x, y, z)$ 在 $P_0(x_0, y_0, z_0)$ 沿 $\vec{l}$ 上方向导数

记为: $\frac{\partial f}{\partial \lambda} \Big|_{P_0} = \frac{\partial f(x_0, y_0, z_0)}{\partial \lambda}$



$$\frac{\partial f}{\partial x} \Big|_{P_0} = e + e = 2e;$$

$$\frac{\partial f}{\partial y} = (x+z)e^{xy} \cdot x, \quad \frac{\partial f}{\partial y} \Big|_{P_0} = e$$

$$\frac{\partial f}{\partial z} = 1 \cdot e^{xy}, \quad \frac{\partial f}{\partial z} \Big|_{P_0} = e. \quad \begin{matrix} P_0(1,1,0) \\ |\vec{l}| = \sqrt{3} \end{matrix}$$

$$\begin{aligned} \therefore \frac{\partial f}{\partial l} \Big|_{P_0} &= \left\{ \frac{\partial f}{\partial x}, \frac{\partial f}{\partial y}, \frac{\partial f}{\partial z} \right\} \Big|_{P_0} \cdot \frac{\vec{l}}{|\vec{l}|} \\ &= \{2e, e, e\} \cdot \frac{\{1, 1, 1\}}{\sqrt{3}} \leftarrow = \frac{4e}{\sqrt{3}}. \end{aligned}$$

给定可微函数 $f(x, y, z)$, 在 $P_0(x_0, y_0, z_0)$ 沿 $\vec{l}$ 方向导数:

$$\vec{l} = \{\cos \alpha, \cos \beta, \cos \gamma\}, \quad |\vec{l}| = 1;$$

$$\begin{aligned} \frac{\partial f}{\partial l} \Big|_{P_0} &= \{f_x(P_0), f_y(P_0), f_z(P_0)\} \cdot \{\cos \alpha, \cos \beta, \cos \gamma\} \\ &= \{f_x(P_0), f_y(P_0), f_z(P_0)\} \cdot \vec{l} \quad \theta \text{ 是夹角} \end{aligned}$$

$$= \|\{f_x(P_0), f_y(P_0), f_z(P_0)\}\| \cdot |\vec{l}| \cdot \cos \theta$$

$$= \sqrt{(f_x(P_0))^2 + (f_y(P_0))^2 + (f_z(P_0))^2} \cdot \cos \theta, \quad |\vec{l}| = 1,$$

$\theta: \cos \theta = 1: \frac{\partial f}{\partial l} \Big|_{P_0} \geq 0$ 最大; $\theta = 0: \vec{l}$ 与 $\{f_x(P_0), f_y(P_0), f_z(P_0)\}$ 同方向.

$\cos \theta = -1: \frac{\partial f}{\partial l} \Big|_{P_0} \leq 0$ 绝对值最大; $\theta = \pi: \vec{l}$ 反方向.

$\cos \theta = 0: \frac{\partial f}{\partial l} \Big|_{P_0} = 0: \text{变化率为零. } \theta = \frac{\pi}{2}, \text{ 垂直.}$

二. 梯度.

定义 2.2 (梯度). $u = f(x, y, z)$ 在 $P_0(x_0, y_0, z_0)$ 可微.

$\{f_x(P_0), f_y(P_0), f_z(P_0)\}$ 称为 $u = f(x, y, z)$ 在 P_0 的梯度, 记为 $\text{grad } f(P_0)$ (gradient: 梯度).

$$\text{或 } \nabla f(P_0) = \nabla f(x_0, y_0, z_0) = \{f_x(P_0), f_y(P_0), f_z(P_0)\} \\ = \text{grad } f(P_0)$$

性质: C : 常数

$$(1) \nabla(C) = \vec{0} = \{0, 0, 0\},$$

$$(2) \nabla(Cu) = C \cdot \nabla u \\ = \{(Cu)_x, (Cu)_y, (Cu)_z\}$$

$$(3) \nabla(f_1 + f_2) = \nabla f_1 + \nabla f_2$$

$$(4) \nabla(u \cdot v) = v \nabla u + u \nabla v$$

$$\triangleq \{(uv)_x, (uv)_y, (uv)_z\}$$

$$= \{u_x v + u v_x, u_y v + u v_y, u_z v + u v_z\}$$

$$= \{u_x v, u_y v, u_z v\} + \{u v_x, u v_y, u v_z\}$$

$$(5) \nabla(f(u)) = f'(u) \cdot \nabla u$$

$$\triangleq \{f(u)_x, f(u)_y, f(u)_z\}$$

$$= \{f'(u) u_x, f'(u) u_y, f'(u) u_z\}$$

线性运算.

Leibniz 法则

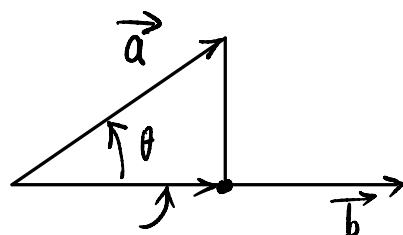
$u=f(x,y,z)$ 可微: $\vec{l}=\{\cos\alpha, \cos\beta, \cos\gamma\}$, $|\vec{l}|=1$,

$$\begin{aligned}\frac{\partial u}{\partial l} &= \{f_x, f_y, f_z\} \cdot \vec{l} = \nabla f \cdot \vec{l} \\ &= \|\nabla f\| \cdot |\vec{l}| \cdot \cos\theta. \quad \theta \text{ 是 } \nabla f \text{ 与 } \vec{l} \text{ 的夹角.} \\ &= \|\nabla f\| \cdot \cos\theta\end{aligned}$$

方向导数 $\frac{\partial u}{\partial l}$ 是

∇f 在 $\vec{l}$ 上的投影

沿 $\vec{l}$ 的变化率.



$\vec{a}$ 在 $\vec{b}$ 上的投影: $|\vec{a} \cdot \vec{b}| = |\vec{a}| \cdot \cos\theta$

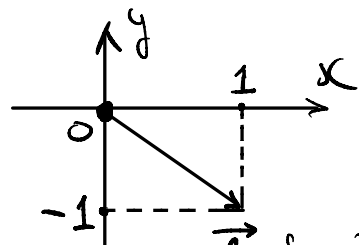
例, 设 $f(x,y)=xe^y$, 求在 $(1,1)$ 处的梯度, 沿 $\vec{l}=\vec{i}-\vec{j}$ 的变化率

解: $f_x = e^y, f_y = xe^y$,

$\vec{l}=\vec{i}-\vec{j}=\{1, -1\}$.

$\therefore f_x(1,1)=e, f_y(1,1)=e,$

$\nabla f(1,1)=\{e, e\},$



$|\vec{l}|=\sqrt{2},$

$\therefore \frac{\partial f}{\partial l} \Big|_{(1,1)} = \nabla f(1,1) \cdot \frac{\vec{l}}{|\vec{l}|}$ 单位向量.

$= \{e, e\} \cdot \frac{\{1, -1\}}{\sqrt{2}}$

$= 0.$ 变化率.

可微 $\Rightarrow$ 偏导数存在

$\Rightarrow$ 沿任何方向方向导数存在.

$$\frac{\partial u}{\partial \vec{r}} \Big|_{P_0} = \nabla f(P_0) \cdot \frac{\vec{r}}{|\vec{r}|} \quad \text{—— 方向导数}$$

证:

反之, 沿任何方向方向导数都存在, ~~⇒~~ 可微.

$$\text{例. } f(x, y) = \begin{cases} \frac{x^2 y}{x^2 + y^2}, & (x, y) \neq (0, 0) \\ 0, & (x, y) = (0, 0) \end{cases}$$

$$0 \leq |f(x, y)| \leq \frac{|x^2 y|}{x^2 + y^2} \leq |y| \leq 1$$

$$0 \leq |f(x, y)| \leq \frac{|x^2 y|}{x^2 + y^2} \leq |y|, \xrightarrow{(x, y) \rightarrow (0, 0)} 0 = f(0, 0)$$

$$\therefore \lim_{(x, y) \rightarrow (0, 0)} f(x, y) = f(0, 0).$$

$\therefore f(x, y)$ 在 $(0, 0)$ 连续.

$$\begin{aligned} \textcircled{2} \quad f_x(0, 0) &= \lim_{x \rightarrow 0} \frac{f(x, 0) - f(0, 0)}{x} = \lim_{x \rightarrow 0} \frac{\frac{x^2 \cdot 0}{x^2 + 0^2} - 0}{x} \\ &= 0 = \frac{d}{dx} f(x, 0) \Big|_{x=0} \end{aligned}$$

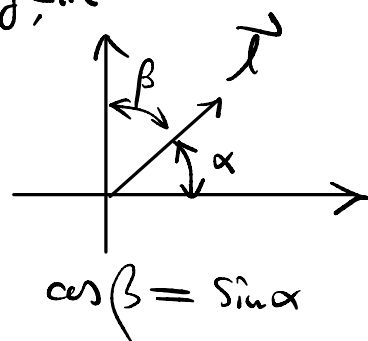
$$y=0: f(x, 0) = \begin{cases} \frac{x^2 \cdot 0}{x^2 + 0^2} \Big|_{y=0}, & x \neq 0 \\ 0, & x=0 \end{cases} = 0, \quad \forall x \in \mathbb{R}.$$

$$\Rightarrow f_x(x, 0) = 0, \quad f_x(0, 0) = 0$$

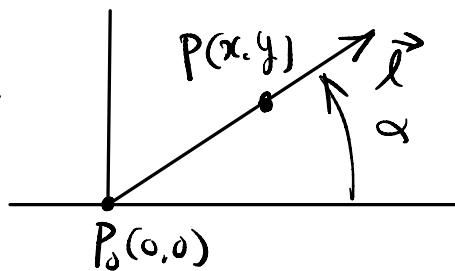
$$f_y(0, 0): f(0, y) = \begin{cases} 0, & y \neq 0 \\ 0, & y=0 \end{cases} = 0, \quad \forall y \in \mathbb{R}$$

$$\Rightarrow f_y(0, y) = 0, \quad f_y(0, 0) = 0,$$

$$\text{对 } \vec{r} = \{\cos \alpha, \cos \beta\} = \{\cos \alpha, \sin \alpha\} \quad 0 \leq \alpha \leq 2\pi.$$



$$P(x, y): \begin{cases} x = t \cos \alpha \\ y = t \sin \alpha \end{cases} \quad t \neq 0:$$



$$f(x, y) = f(t \cos \alpha, t \sin \alpha)$$

$$= \frac{(t \cos \alpha)^2 + t \sin \alpha}{(t \cos \alpha)^2 + (t \sin \alpha)^2}$$

$$\frac{x^2 y}{x^2 + y^2}$$

$$= t \cos^2 \alpha \sin \alpha$$

$$\left. \frac{\partial f}{\partial t} \right|_{(0,0)} = \lim_{t \rightarrow 0} \frac{f(x, y) - f(0,0)}{t} = \lim_{t \rightarrow 0} \frac{t \cos^2 \alpha \sin \alpha}{t} = \cos^2 \alpha \sin \alpha$$

$$\therefore \left. \frac{\partial f}{\partial t} \right|_{(0,0)} = \cos^2 \alpha \sin \alpha$$

③ 但: 倘若 $f(x, y)$ 在 $(0, 0)$ 可微

$$\rho = \sqrt{x^2 + y^2} \rightarrow 0:$$

$$\Rightarrow f(x, y) - f(0, 0) = \left. \frac{\partial f}{\partial x} \right|_{(0,0)} x + \left. \frac{\partial f}{\partial y} \right|_{(0,0)} y + o(\rho)$$

o(\rho) 的阶数

$$\text{这时有} \quad f(x, y) = o(\rho).$$

$$\Leftrightarrow \frac{f(x, y)}{\rho} \rightarrow 0 \quad (\rho \rightarrow 0)$$

$$\text{例: } \frac{f(x, y)}{\rho} = \frac{\frac{x^2 y}{x^2 + y^2}}{\rho} = \frac{x^2 y}{\rho^3}$$

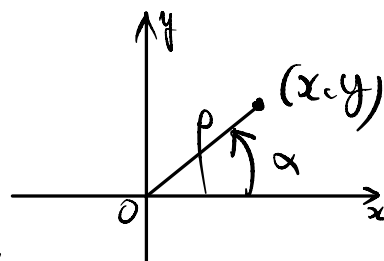
$$= \frac{(p \cos \alpha)^2 \cdot p \sin \alpha}{p^3}$$

$$\begin{cases} x = p \cos \alpha \\ y = p \sin \alpha \end{cases}$$

$$= \cos^2 \alpha \sin \alpha \neq 0$$

$$\alpha \neq 0, \pi: \frac{\pi}{2},$$

$$\frac{f(x, y)}{\rho} \not\rightarrow 0 \quad (\rho \rightarrow 0)$$



$\therefore f(x, y)$ 在 $(0, 0)$ 不可微!

更有甚者:

$\exists f(x, y)$: 在 (x_0, y_0) 沿任意方向方向导数存在
且相等. ~~$\Rightarrow$ 连续~~

例. 设 $f(x, y) = \begin{cases} 1, & 0 < y < x^2 \\ 0, & y \leq 0 \text{ 或 } y \geq x^2 \end{cases}$

$E = \{(x, y) \mid 0 < y < x^2\}$:

$f(x, y) = \chi_E(x, y)$.

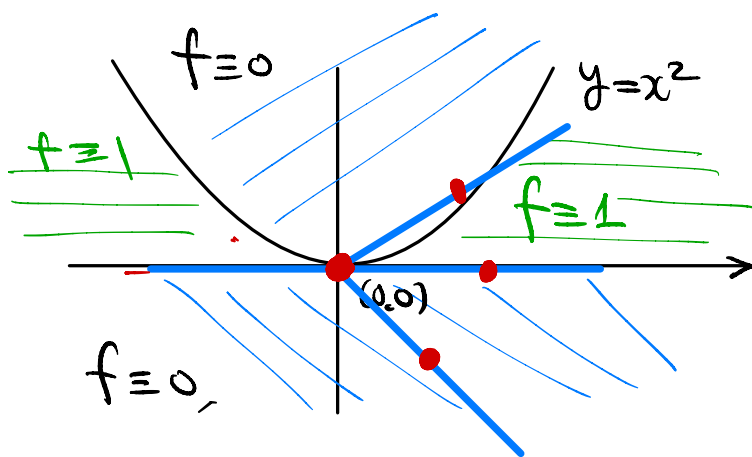
在 $(0, 0)$ 不连续! $\checkmark$

$\forall \vec{l} = \{\cos \alpha, \cos \beta\}$:

$$\left. \frac{\partial f}{\partial \vec{l}} \right|_{(0,0)} = 0. \quad \checkmark \quad (\text{Ex})$$

即沿任何方向, 方向导数存在且相等!

尽管如此, $f(x, y)$ 在 $(0, 0)$ 不连续!



作业. P137. 1. 2. 3. 4. 5.